# Lecture 14 – Introduction to hypothesis testing

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Hypothesis testing answers binary questions about a parameter, such as...

- Our first example from class: do people prefer to turn their head to the right when kissing?
- Is a drug better than placebo?
- Is the average waiting time in the ER $> 30$ minutes?
- Is my data Gaussian?

How do we formulate such a question using statistics?

Consider the ER question. Suppose we collect i.i.d. data $X_1, \ldots, X_n$, the ER waiting times of different patients. Our parameter of interest is $\mu = \mathbb{E}[X_1]$. The binary question about the parameter is: is $\mu \leq 30$ or is $\mu > 30$?

How do we go about answering this question?

The first strategy that comes to mind is to compute $\bar{X}_n$, and check if $\bar{X}_n > 30$. But $\bar{X}_n$ is not a perfect proxy for $\mu$. We could have $\bar{X}_n > 30$ even if $\mu < 30$. So we should only decide $\mu > 30$ if $\bar{X}_n > 30 + c$ for a large enough $c$. For example if $\bar{X}_n > 60$ minutes, then $\mu < 30$ is very unlikely (but not impossible). With incomplete data we have to accept the possibility of error. The language of hypothesis testing will help us reason about the types of errors and probability of making them, and how we should decide where to set the cut off.

## 14.1 Null and alternative hypotheses

**Definition 1** (Hypotheses)**.** In a statistical model $\{f_\theta : \theta \in \Theta\}$, a **hypothesis** is a statement that $\theta$ belongs to a specified subset of $\Theta$. We consider

$$\underbrace{H_0 : \theta \in \Theta_0}_{\text{null hypothesis}} \quad \text{vs} \quad \underbrace{H_1 : \theta \in \Theta_1}_{\text{alternative hypothesis}},$$

where $\Theta_0 \cap \Theta_1 = \varnothing$ and $\Theta_0 \cup \Theta_1 \subseteq \Theta$.

**Example 1.** In the ER waiting time example, we might test

$$H_0 : \mu \leq 30 \quad \text{vs} \quad H_1 : \mu > 30.$$

For the effectiveness of a drug regulating blood pressure, let $\theta$ be the average change in a patient's blood pressure after taking the drug. We might test

$$\underbrace{H_0 : \theta = 0}_{\text{No effect}} \quad \text{vs} \quad \underbrace{H_1 : \theta \neq 0}_{\text{Presence of effect}}$$

How do we choose which hypothesis to call the null ($H_0$) and which to call the alternative ($H_1$)? The operating principles are

$$\textit{innocent } (H_0) \textit{ until proven guilty } (H_1),$$

or

$$\textit{status quo } (H_0) \textit{ vs discovery } (H_1).$$

**Example 2.**

- Suppose someone is suing Duke hospital for falsely claiming that their waiting times are below 30 minutes. The default presumption is $H_0 : \mu \leq 30$ (hospital is innocent). To prove the alternative hypothesis $H_1 : \mu > 30$ (guilty), the person suing the hospital would need to bring data as evidence to reject the default assumption of innocence.
- A pharma company petitions the FDA to approve the drug they developed. Then $H_0$ : the placebo outperforms the drug (status quo), $H_1$: the drug outperforms the placebo (a scientific discovery).
- A scientist claims she's discovered a gene that cures all cancer. Then $H_0$ : cancer-curing gene doesn't exist (status quo), $H_1$ : cancer-curing gene really does exist (scientific discovery) .

*The alternative is the claim for which we require convincing evidence.*

**Remark 1** (Interpreting a decision)**.** We either **reject** $H_0$ or **fail to reject** $H_0$ (rather than "accept $H_1$"). Failing to reject means only that there is insufficient evidence against the null.

## 14.2 Hypothesis tests

**Definition 2.** A **hypothesis test** is a rule that specifies

1. For which sample values the decision is to reject $H_0$
2. For which sample values the decision is to not reject $H_0$.

A hypothesis test can be specified by a function $\psi \colon \mathcal{X}^n \to \{0, 1\}$ where

$$\begin{aligned} \psi(x) = 1 \quad &\implies \quad H_0 \text{ is rejected} \\ \psi(x) = 0 \quad &\implies \quad H_0 \text{ is not rejected} \end{aligned}$$

This function partitions the sample space $\mathcal{X}^n$ into two disjoint sets:

$$\begin{aligned} \mathcal{R} &\coloneqq \{x : \psi(x) = 1\} \qquad \text{(Rejection region)} \\ \mathcal{R}^c &\coloneqq \{x : \psi(x) = 0\} \qquad \text{(Region of failing to reject)} \end{aligned}$$

In many cases, the function $\psi$ is defined in terms of some statistic $T$.

**Example 3.** Continuing with the ER waiting times question, recall the hypotheses

$$H_0 : \mu \leq 30 \quad \text{versus} \quad H_1 : \mu > 30.$$

As we discussed, a natural approach is to reject based on the value of $T(\underline{x}) = \frac{1}{n}\sum_{i=1}^n x_i$. For example, we might choose

$$\psi(\underline{x}) = \mathbb{1}_{\left\{\frac{1}{n}\sum_{i=1}^n x_i > 31\right\}} \quad \iff \quad \mathcal{R} = \left\{\underline{x} : \frac{1}{n}\sum_{i=1}^n x_i > 31\right\}$$

Is this test working well? To evaluate this, we need to talk about error types.

## 14.3 Error types

If we are performing a test, then we do not know whether $H_0$ or $H_1$ is the true one, i.e. whether the true parameter is in $\Theta_0$ or $\Theta_1$! So the types of errors we can make depend on the truth and our decision:

| | | **Decision of Test** | |
|---|---|---|---|
| | | $H_0$ $(\psi = 0)$ | $H_1$ $(\psi = 1)$ |
| **Truth** | $H_0$ | Correct | Type I Error |
| | $H_1$ | Type II Error | Correct |

By the "innocent until proven guilty" principle, Type I is the more serious error.

**Example 4.** A test commits a Type I error if it concludes a drug works better than the placebo when it actually doesn't. This is considered a more serious error than a Type II error, in which the test concludes the drug does not work better than placebo, when it actually does.

**Error probabilities:**

$$\begin{aligned} \text{Probability of Type I error :} \quad & \mathbb{P}_\theta(\psi(\underline{X}) = 1), \quad \theta \in H_0, \\ \text{Probability of Type II error :} \quad & \mathbb{P}_\theta(\psi(\underline{X}) = 0), \quad \theta \in H_1. \end{aligned}$$

Note that the probability of the two error types depends not just on whether $H_0$ or $H_1$ is true, but on the specific $\theta$ value within $H_0$ or $H_1$.

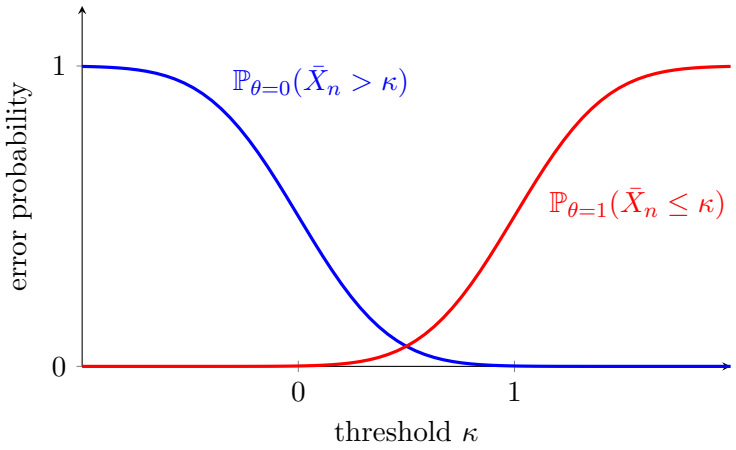
**Example 5** (Tradeoff between error types)**.** Suppose $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\theta, 1)$. Consider the hypotheses

$$H_0 : \theta = 0 \quad \text{versus} \quad H_1 : \theta = 1.$$

We'll use the test $\psi(\underline{x}) = \mathbb{1}_{\{\bar{x}_n > \kappa\}}$ for some $\kappa$. How should we choose $\kappa$? Consider what happens to the Type I and Type II errors. They are:

$$\begin{aligned} \mathbb{P}(\text{Type I}) &= \mathbb{P}_{\theta=0}(\bar{X}_n > \kappa), \quad \text{(decreasing in } \kappa) \\ \mathbb{P}(\text{Type I}) &= \mathbb{P}_{\theta=1}(\bar{X}_n \leq \kappa), \quad \text{(increasing in } \kappa) \end{aligned}$$

Thus, there is a tradeoff between the probabilities of Type I error and Type II errors. It is impossible to make both error probabilities arbitrarily small.




**Definition 3.** Let $\psi(\underline{x})$ be a hypothesis test.

- $\psi$ is a test of **size** $\alpha$ if

$$\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(\psi(\underline{X}) = 1) = \alpha.$$

- $\psi$ is a test of **level** $\alpha$ if

$$\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(\psi(\underline{X}) = 1) \leq \alpha.$$

A test having level $\alpha$ means that the probability of Type I error is no greater than $\alpha$.

**Strategy for designing tests:** Typically, when we design a hypothesis test, we specify in advance that its level should be no more than $\alpha$; for example, $\alpha = 1\%$ and $\alpha = 5\%$ are standard. This is the max type I error we are willing to tolerate. Of these level $\alpha$ tests, which should we pick? For example, why not just choose the test $\psi$ which always sets $\psi(\underline{x}) = 0$? It has size zero and is a valid level $\alpha = 1\%$ test. It never commits a type I error! Such a test says "everyone is innocent no matter what", or "no drug works better than placebo". Clearly, such a test is not very useful, it doesn't lead to any discoveries.

Therefore, given a certain allowable level $\alpha$, we try to max out the Type I error over all $\theta \in \Theta_0$. As we will see, doing so helps keep the Type II error low.

## 14.4 Power Function

**Definition 4.** Given a test for $H_0 \colon \theta \in \Theta_0$ versus $H_1 \colon \theta \in \Theta_1$, the **power function** $\beta \colon \Theta \to [0, 1]$ is defined by

$$\beta(\theta) = \mathbb{P}_\theta(\psi = 1).$$

This is the probability of rejecting $H_0$ when $\theta$ is the true parameter.

If the hypotheses are *simple*, i.e.,

$$H_0 : \theta = \theta_0 \quad \text{versus} \quad H_1 : \theta = \theta_1,$$

then

$$\beta(\theta_0) = \text{Probability of Type } I \text{ error},$$
$$\beta(\theta_1) = 1 - \text{Probability of Type } II \text{ error}.$$

In general, we want $\beta(\theta)$ to be small when $\theta \in \Theta_0$ and large when $\theta \in \Theta_1$.

For a perfect test $\psi$ (see Figure 14.1) the power function $\beta$ is a step function: it is exactly zero when $\theta \in \Theta_0$, and exactly one when $\theta \in \Theta_1$.

The power functions in Figures 14.2 and 14.3 both correspond to tests with level $\alpha$. But the former is more efficient, because it maxes out the Type I error at exactly $\alpha$ when crossing over from $\Theta_0$ to $\Theta_1$.

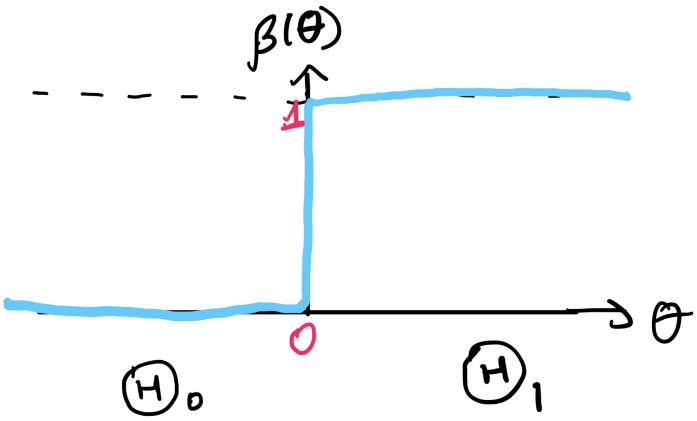



Figure 14.1: The power function for a perfect test, with zero type I error and zero type II error.

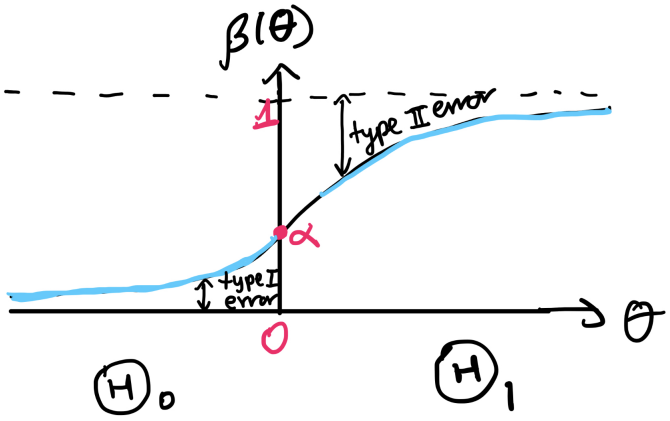



Figure 14.2: The power function for a test of size $\alpha$ and level $\alpha$. Note the largest type I error is achieved at the boundary between $\Theta_0$ and $\Theta_1$.

## 14.5 Worked example

Return to the ER question, $H_0 : \mu \leq 30$ versus $H_1 : \mu > 30$. Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2), \qquad \sigma > 0 \text{ known}.$$

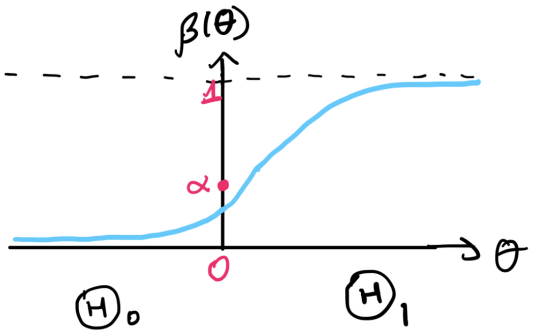



Figure 14.3: The power function for a level $\alpha$ test, but whose size is less than $\alpha$.

Then $\bar{X}_n \sim \mathcal{N}(\mu, \sigma^2/n)$ exactly. Consider a test of the form $\psi(\underline{X}) = \mathbb{1}_{\{\bar{X}_n > 30+c\}}$. We want to choose $c$ to make this exactly a size $\alpha$ test. We can express the size in terms of the power function:

$$\text{size}(\psi) = \sup_{\mu \le 30} \beta(\mu),$$

where recall that $\beta(\mu)$ is the probability of rejecting the null when the true mean is $\mu$. Therefore

$$\begin{aligned}
\beta(\mu) &= \mathbb{P}_\mu(\bar{X}_n > 30 + c) \\
&= \mathbb{P}\left(\mu + \frac{\sigma}{\sqrt{n}} Z > 30 + c\right) \\
&= \mathbb{P}\left(Z > \frac{30 + c - \mu}{\sigma/\sqrt{n}}\right) = 1 - \Phi\left(\frac{30 + c - \mu}{\sigma/\sqrt{n}}\right), \qquad Z \sim \mathcal{N}(0,1).
\end{aligned}$$

As $\mu$ increases, the cutoff inside $\Phi$ decreases, so $\beta(\mu)$ increases. This proves that the largest Type I error possible under the null is at $\mu = 30$:

$$\text{size}(\psi) = \sup_{\mu \le 30} \beta(\mu) = \beta(30) = 1 - \Phi(c\sqrt{n}/\sigma).$$

We want to set this equal to $\alpha$. Define the **upper-tail normal quantile**

$$z_\alpha = \Phi^{-1}(1 - \alpha), \qquad \mathbb{P}(Z > z_\alpha) = \alpha.$$

Then we should set $c\sqrt{n}/\sigma = z_\alpha$, or $c = z_\alpha \sigma/\sqrt{n}$. Putting it all together, the size $\alpha$ test makes the following decision:

$$\text{Reject} \quad \text{if} \quad \bar{X}_n > 30 + \frac{\sigma}{\sqrt{n}} z_\alpha.$$

**Example 6** (Applying the test)**.** Suppose $n = 164$, $\sigma = 12$ minutes, and the observed mean is $\bar{x}_n = 33.4$. We have $z_{0.05} = 1.645$, so

$$30 + z_{0.05}\frac{\sigma}{\sqrt{n}} = 30 + 1.645\frac{12}{\sqrt{164}} \approx 31.54.$$

We reject when $\bar{x}_n > 31.54$. Since $33.4 > 31.54$, we reject $H_0$: the data provide evidence that the population mean exceeds 30 minutes.